

Model Name: P550HVN09.1

Issue Date : 2018/01/30

(*) Preliminary Specifications

() Final Specifications

Customer Signature	Date	AUO	Date
Approved By _____		Approval By PM Director CTW _____	
Note		Reviewed By RD Director 陳品川 _____	
		Reviewed By Project Leader 吳國文 _____	
		Prepared By PM Silva Chiu _____	

Contents

1. General Description	4
1.1. Display Characteristics	4
1.2. Optical Characteristics	5
1.3. Mechanical Characteristics	8
1.3.1. Placement Suggestions	8
2. Absolute Maximum Ratings	11
3. Electrical Specification	12
3.1. Electrical Characteristics	12
3.1.1. DC Characteristics ($T_a = 25 \pm 2^\circ\text{C}$)	12
3.2. Interface Connections	14
3.3. Input Data Format	17
3.3.1. Data mapping	17
3.3.2. Color Input Data Reference	18
3.4. Signal Timing Specification	19
3.5. Input interface characteristics	21
3.5.1. LVDS	21
3.6. Power Sequence for LCD	24
3.7. Backlight Specification	25
3.7.1. Electrical specification ($T_a = 25 \pm 2^\circ\text{C}$)	25
3.7.2. Input Pin Assignment	26
3.7.3. Power Sequence for Backlight	26
3.7.4. LED Operating Life Time	29
4. Reliability Test Items	30
5. International Standard	31
5.1. Safety	31
5.2. EMC	31
6. Packing	32
6.1. Definition of Label	32
6.2. Packing Methods	33
6.3. Pallet and Shipment Information	34
7. Precautions	35
7.1. Mounting Precautions	35
7.2. Operating Precautions	35
7.3. Operating Condition for Public Information Display	36
7.4. Electrostatic Discharge Control	36
7.5. Precautions for Strong Light Exposure	36
7.6. Storage	37
7.7. Handling Precautions for Protection Film	37

[illegible]

1. General Description

This specification applies to the 55 inch Color TFT-LCD Module P550HVN09.1 This LCD module has a TFT active matrix type liquid crystal panel 1920x1080 pixels, and diagonal size of 55 inch. This module supports 1920x1080 mode. Each pixel is divided into Red, Green and Blue sub-pixels or dots which are arranged in vertical stripes. Gray scale or the brightness of the sub-pixel color is determined with a 10-bit (8bit + FRC) gray scale signal for each dot.

The P550HVN09.1 has been designed to apply the 10-bit selectable 2 channel LVDS interface method. It is intended to support displays where high brightness, wide viewing angle.

* General Information

1.1. Display Characteristics

Items	Specification	Unit	Note
Active Screen Size	55	inch	
Display Area	1209.6 (H) x 680.4 (V)	mm	
Outline Dimension	1231.4 (H) x 702.2 (V) x 10.1 (D Min.)	mm	D Min.: Front bezel to Back Bezel
Driver Element	a-Si TFT active matrix		
Bezel Opening	1213.6 (H) x 684.4 (V)	mm	
Display Colors	1073M	Colors	
Number of Pixels	1920x1080	Pixel	
Pixel Pitch	0.63 (H) x 0.63 (W)	mm	
Pixel Arrangement	RGB vertical stripe		
Display Operation Mode	Normally Black		
Surface Treatment	Anti-Glare, 3H		Haze=2%
Rotate Function	Unachievable		Note 1
Display Orientation	Portrait/Landscape Enabled		Note 2

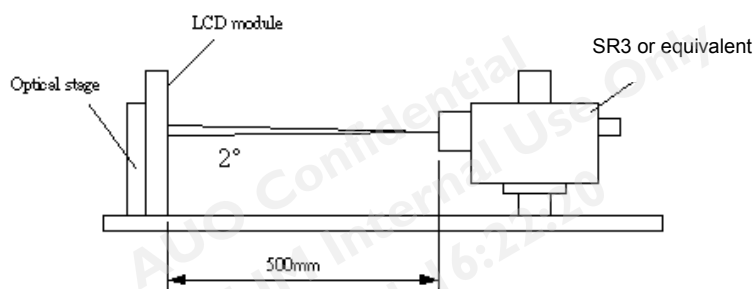
Note 1: Rotate Function refers to LCD display could be able to rotate. This function does not work in this model.

Note 2: Please refer to 1.3.1 Placement Suggestions.

1.2. Optical Characteristics

Optical characteristics are determined on the back-light of measured unit is 'ON' and stabilized after 45~60 minutes in a dark environment at 25°C. The values are specified at 50cm distance from the LCD surface at a viewing angle of φ and θ equal to 0°.

Fig.1 presents additional information concerning the measurement equipment and method.



Parameter		Symbol	Values			Unit	Notes
			Min.	Typ.	Max		
Contrast Ratio		CR	3200	4000	--		1
Surface Luminance (White)		$L_{WH}(2D)$	360	450	--	cd/m ²	2
Luminance Variation		$\delta_{WHITE(9P)}$	--	--	1.33		3
Response Time (G to G)		T _y	--	8	--	ms	4
Color Gamut		NTSC		72		%	
Color Coordinates							
	Red	R _x	Typ.-0.03	0.648	Typ.+0.03		
		R _y		0.337			
	Green	G _x		0.313			
		G _y		0.614			
	Blue	B _x		0.152			
		B _y		0.062			
	White	W _x		0.280			
		W _y		0.290			
Viewing Angle							5
	x axis, right($\varphi=0^\circ$)	θ_r	--	89	--	degree	
	x axis, left($\varphi=180^\circ$)	θ_l	--	89	--	degree	
	y axis, up($\varphi=90^\circ$)	θ_u	--	89	--	degree	
	y axis, down ($\varphi=270^\circ$)	θ_d	--	89	--	degree	

Note:

1. Contrast Ratio (CR) is defined mathematically as:

$$\text{Contrast Ratio} = \frac{\text{Surface Luminance of } L_{\text{on5}}}{\text{Surface Luminance of } L_{\text{off5}}}$$

2. Surface luminance is luminance value at point 5 across the LCD surface 50cm from the surface with all pixels displaying white. From more information see FIG 2. LED input $V_{\text{DDB}} = 24\text{V}$, $I_{\text{DDB}} = 2.54\text{A}$, $L_{\text{WH}} = L_{\text{on5}}$ where L_{on5} is the luminance with all pixels displaying white at center 5 location.

3. The variation in surface luminance, δWHITE is defined (center of Screen) as:

$$\delta\text{WHITE(9P)} = \text{Maximum}(L_{\text{on1}}, L_{\text{on2}}, \dots, L_{\text{on9}}) / \text{Minimum}(L_{\text{on1}}, L_{\text{on2}}, \dots, L_{\text{on9}})$$

4. Response time T_r is the average time required for display transition by switching the input signal for five luminance ratio (0%,25%,50%,75%,100% brightness matrix) and is based on $F_v = 60\text{Hz}$ to optimize.

Measured		Target				
Response Time		0%	25%	50%	75%	100%
Start	0%		0% to 25%	0% to 50%	0% to 75%	0% to 100%
	25%	25% to 0%		25% to 50%	25% to 75%	25% to 100%
	50%	50% to 0%	50% to 25%		50% to 75%	50% to 100%
	75%	75% to 0%	75% to 25%	75% to 50%		75% to 100%
	100%	100% to 0%	100% to 25%	100% to 50%	100% to 75%	

T_r is determined by 10% to 90% brightness difference of rising or falling period. (As illustrated)

The response time is defined as the following figure and shall be measured by switching the input signal for “any level of gray(bright)” and “any level of gray(dark)”.

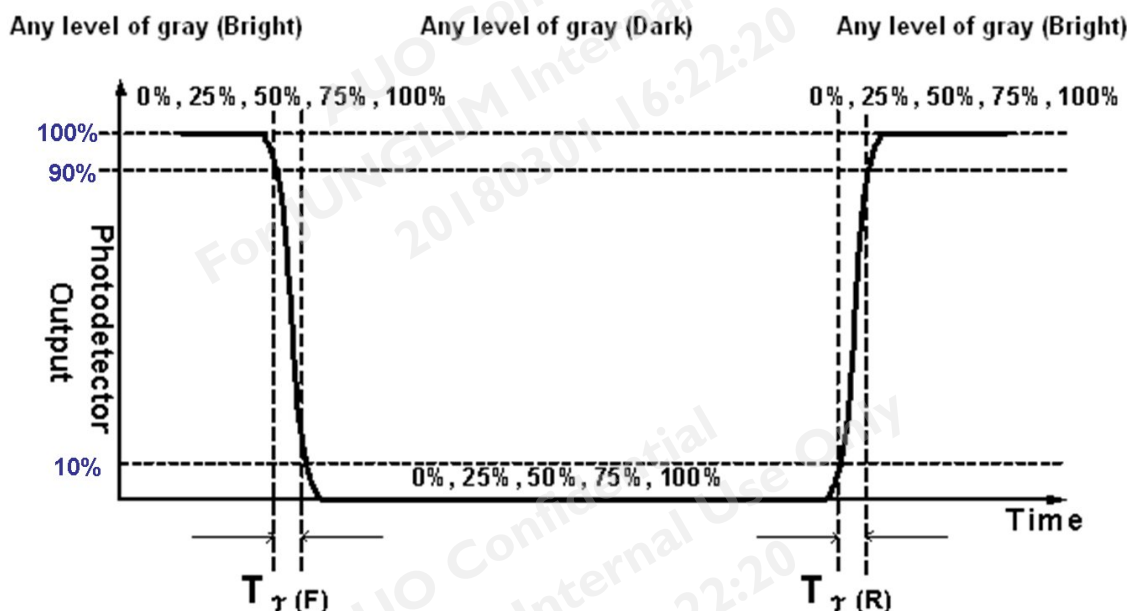
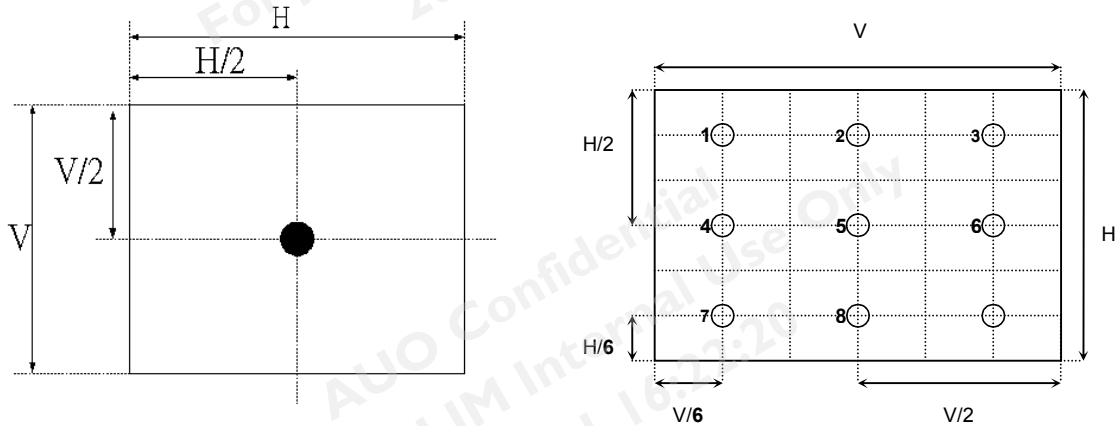
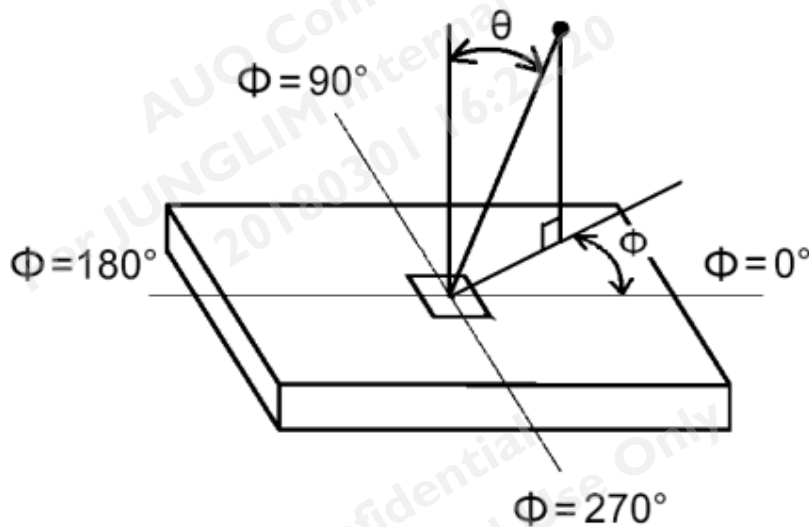


FIG. 2 Luminance



5. Viewing angle is the angle at which the contrast ratio is greater than 10. The angles are determined for the horizontal or x axis and the vertical or y axis with respect to the z axis which is normal to the LCD surface. For more information see FIG3.

FIG.3 Viewing Angle



1.3. Mechanical Characteristics

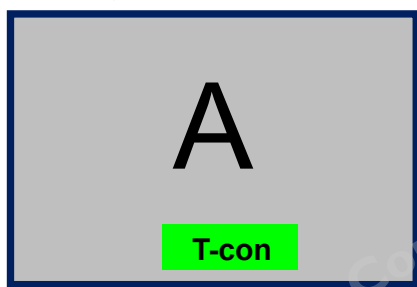
The contents provide general mechanical characteristics for the model P550HVN09.1 In addition the figures in the next page are detailed mechanical drawing of the LCD.

Item		Dimension	Unit	Note
Outline Dimension	Horizontal	1231.4	mm	
	Vertical	702.2	mm	
	Depth (Dmin)	10.1	mm	front bezel to back bezel
	Depth (Dmax)	44.45	mm	to stud
Weight	13200 (Typ)		g	w/ DB

1.3.1. Placement Suggestions

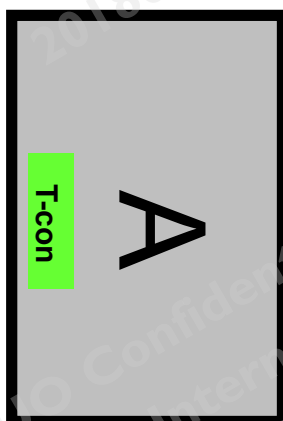
1. Landscape Mode: The default placement is T-Con Side on the lower side and the image is shown upright via viewing from the front.

(Front View)

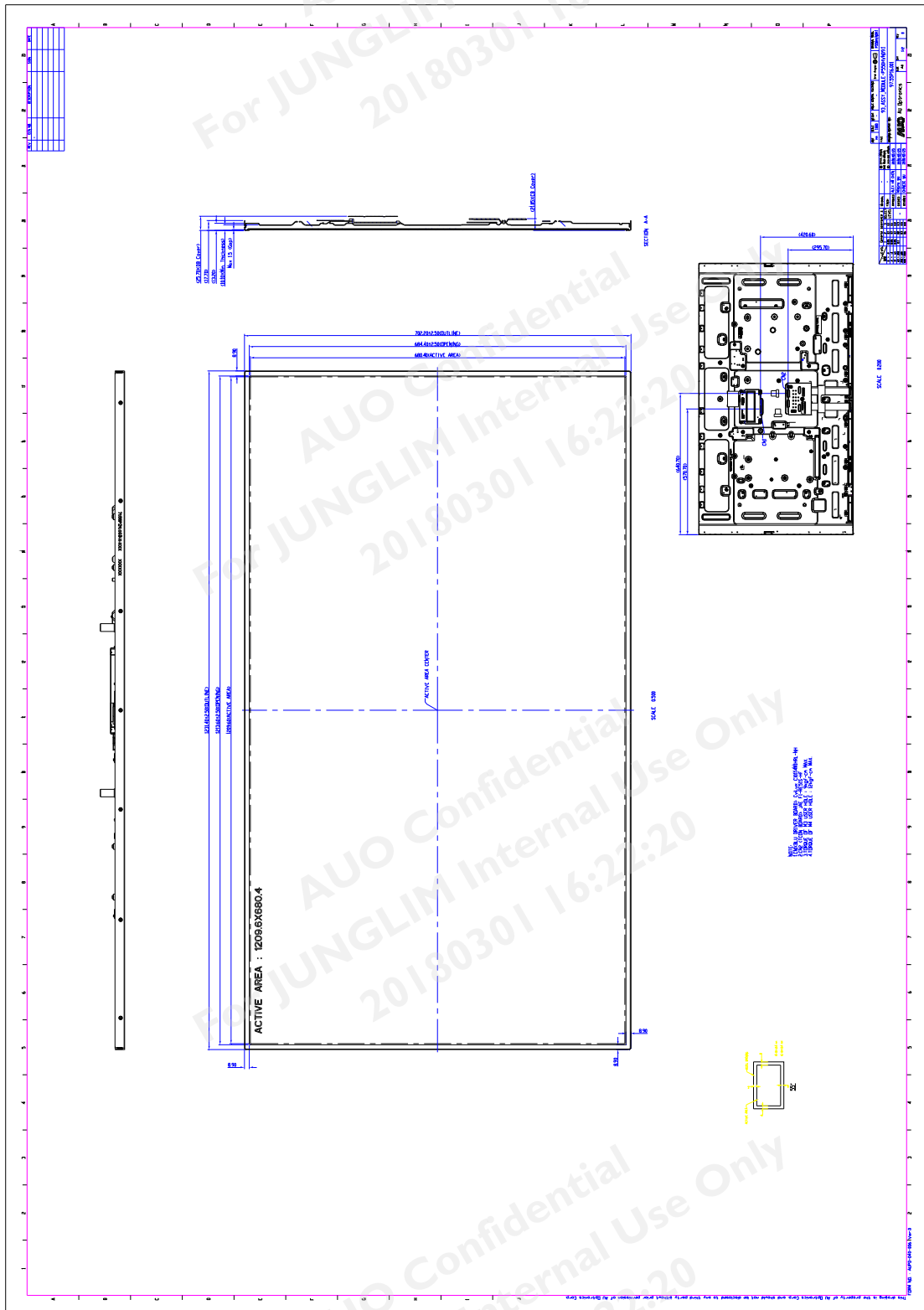


2. Portrait Mode: The default placement is that T-Con side has to be placed on the left side via viewing from the front.

(Front View)



Front View



For JUNGLE 20180301

Confidential Use Only

19.00
482.6

17.00
431.8

12.00
304.8

1.00
25.4

0.50
12.7

0.25
6.35

0.125
3.175

0.0625
1.5875

0.03125
0.79375

0.015625
0.396875

0.0078125
0.1984375

0.00390625
0.09921875

0.001953125
0.049609375

0.0009765625
0.0248046875

0.00048828125
0.01240234375

0.000244140625
0.006201171875

0.0001220703125
0.0031005859375

0.00006103515625
0.00155029296875

0.000030517578125
0.000775146484375

0.0000152587890625
0.0003875732421875

0.00000762939453125
0.00019378662109375

0.000003814697265625
0.000096893310546875

0.0000019073486328125
0.0000484466552734375

0.00000095367431640625
0.00002422332763671875

0.000000476837158203125
0.000012111663818359375

0.0000002384185791015625
0.0000060558319091796875

0.00000011920928955078125
0.00000302791595458984375

0.000000059604644775390625
0.000001513957977294921875

0.0000000298023223876953125
0.0000007569789886474609375

0.00000001490116119384765625
0.00000037848949432373046875

0.000000007450580596923828125
0.000000189244747161865234375

0.0000000037252902984619140625
0.0000000946223735809326171875

0.00000000186264514923095703125
0.00000004731118679046630859375

0.000000000931322574615478515625
0.000000023655593395233154296875

0.0000000004656612873077392578125
0.0000000118277966976165771484375

0.00000000023283064365386962890625
0.00000000591389834880828857421875

0.000000000116415321826934814453125
0.000000002956949174404144287109375

0.0000000000582076609134674072265625
0.0000000014784745872020721435546875

0.00000000002910383045673370361328125
0.00000000073923729360103607177734375

0.000000000014551915228366851806640625
0.000000000369618646800518035888671875

0.0000000000072759576141834259033203125
0.0000000001848093234002590179443359375

0.00000000000363797880709171295166015625
0.00000000009240466170012950897216796875

0.000000000001818989403545856475830078125
0.000000000046202330850064754486083984375

0.0000000000009094947017729282379150390625
0.0000000000231011654250323772430419921875

0.00000000000045474735088646411895751953125
0.00000000001155058271251618862152099609375

0.000000000000227373675443232059478759765625
0.000000000005775291356258094310760498046875

0.0000000000001136868377216160297393798828125
0.0000000000028876456781290471553802490234375

0.00000000000005684341886080801486968994140625
0.00000000000144382283906452357769012451171875

0.000000000000028421709430404007434844970703125
0.000000000000721911419532261788845062255859375

0.0000000000000142108547152020037174224883515625
0.0000000000003609557097661308944225311279296875

0.00000000000000710542735760100185871124417578125
0.00000000000018047785488306544721126570896484375

0.000000000000003552713678800500929355622087890625
0.000000000000090238927441532723605632854482421875

0.0000000000000017763568394002504646778110439453125
0.00000000000004511946372076636180281642724121875

0.00000000000000088817841970012523223890552197265625
0.0000000000000225597318603831809014072136206171875

0.000000000000000444089209850062616119452760986328125
0.00000000000001127986593019159045070360681030859375

0.0000000000000002220446049250313080597263804931640625
0.0000000000000056399329650957952253518034051524296875

0.00000000000000011102230246251565402986319024658203125
0.00000000000000281996648254789761267590170257621484375

0.000000000000000055511151231257827014931595123291015625
0.0000000000000014099832412739488063379508512645621875

0.0000000000000000277555756156289135074657975616455078125
0.00000000000000070499162063697440316897502530228125

0.00000000000000001387778780781445675373289878082275390625
0.00000000000000035249581031848720158448751265114296875

0.000

2. Absolute Maximum Ratings

The followings are maximum values which, if exceeded, may cause faulty operation or damage to the unit

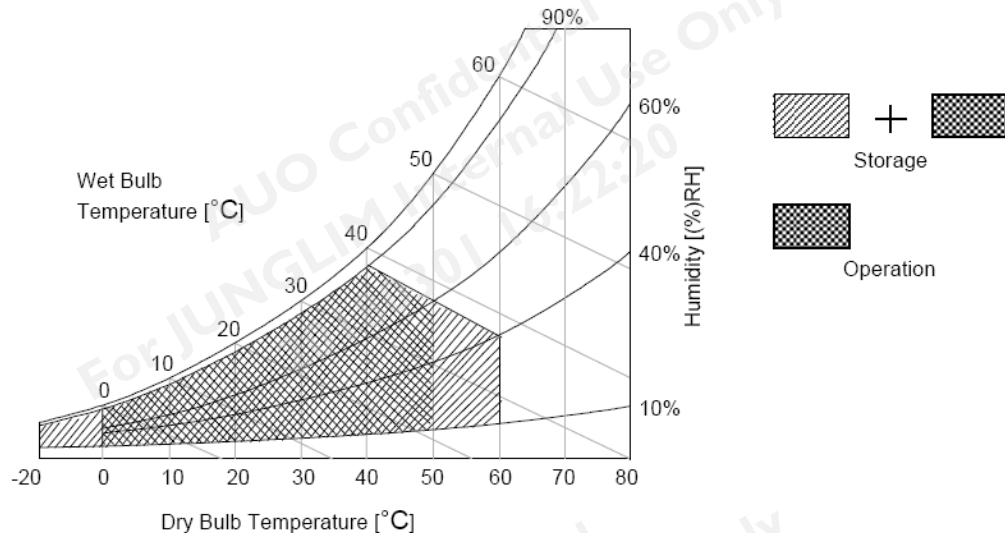
Item	Symbol	Min	Max	Unit	Conditions
Logic/LCD Drive Voltage	Vcc	-0.3	14	[Volt]	Note 1
Input Voltage of Signal	Vin	-0.3	4	[Volt]	Note 1
Operating Temperature	TOP	0	+50	[°C]	Note 2
Operating Humidity	HOP	10	90	[%RH]	Note 2
Storage Temperature	TST	-20	+60	[°C]	Note 2
Storage Humidity	HST	10	90	[%RH]	Note 2
Panel Surface Temperature	PST		65	[°C]	Note 3

Note 1: Duration:50 msec.

Note 2 : Maximum Wet-Bulb should be 39°C and No condensation.

The relative humidity must not exceed 90% non-condensing at temperatures of 40°C or less. At temperatures greater than 40°C, the wet bulb temperature must not exceed 39°C.

Note 3: Surface temperature is measured at 50°C Dry condition



3. Electrical Specification

The P550HVF12.0 requires two power inputs. One is employed to power the LCD electronics and to drive the TFT array and liquid crystal. The other is to power Back Light Unit.

3.1. Electrical Characteristics

3.1.1. DC Characteristics (Ta = 25 ± 2 °C)

Parameter		Symbol	Value			Unit	Note
			Min.	Typ.	Max		
LCD							
Power Supply Input Voltage		V _{DD}	10.8	12	13.2	V _{DC}	1
Power Supply Input Current	Black pattern	I _{DD}	--	0.48	0.64	A	2
	White pattern		--	1.26	1.63	A	
	H-strip pattern		--	0.85	1.14	A	
Power Consumption	Black pattern	P _C	--	5.76	7.68	Watt	
	White pattern		--	15.12	19.56	Watt	
	H-strip pattern		--	10.2	13.68	Watt	
Backlight Power Consumption		P _{BL}		61	66.3	Watt	
Inrush Current		I _{RUSH}	--	--	5.7	A	3
Life time (MTTF)			30000			Hour	4, 5

Note1. The ripple voltage should be fewer than 5% of VDD.

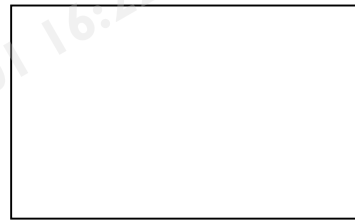
Note2. Test Condition:

- (1) V_{DD} = 12.0V, (2) Fv = 60Hz, (3) Fclk = 74.25MHz, (4) Temperature = 25 °C
 (5) Power dissipation check pattern. (Only for power design)

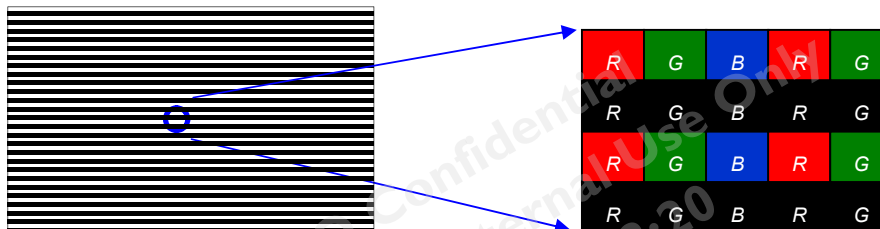
a. Black pattern



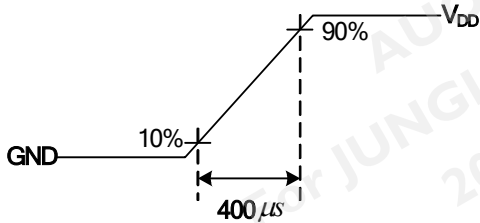
b. White pattern



c. H-Strip pattern



Note3. Measurement condition : Rising time = 400us



Note4. The relative humidity must not exceed 80% non-condensing at temperatures of $40^{\circ}C$ or less. At temperatures greater than $40^{\circ}C$, the wet bulb temperature must not exceed $39^{\circ}C$.

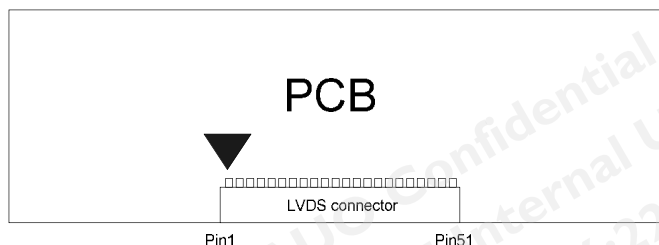
Note5. The lifetime (MTTF) is defined as the time which luminance of LED is 50% compared to its original value.
[Operating condition: Continuous operating at $T_a = 25 \pm 2^{\circ}C$, for single lamp/LED only]

3.2. Interface Connections

- LCD connector: JAE FI-RE51S-HF (LVDS connector) or compatible

PIN	Symbol	Description	Note	PIN	Symbol	Description	Note
1	N.C.	No connection	2	26	GND	Ground	
2	SCL	I2C Clock	3,4	27	GND	Ground	
3	WP	Write Protection	3,5	28	CH2_0-	LVDS Channel 2, Signal 0-	
4	SDA	I2C Data	3,4	29	CH2_0+	LVDS Channel 2, Signal 0+	
5	N.C.	No connection		30	CH2_1-	LVDS Channel 2, Signal 1-	
6	N.C.	No connection	2	31	CH2_1+	LVDS Channel 2, Signal 1+	
7	LVDS_SEL	LVDS data format selection	3,6	32	CH2_2-	LVDS Channel 2, Signal 2-	
8	N.C.	No connection	2	33	CH2_2+	LVDS Channel 2, Signal 2+	
9	N.C.	No connection	2	34	GND	Ground	
10	N.C.	No connection	2	35	CH2_CLK-	LVDS Channel 2, Clock -	
11	GND	Ground		36	CH2_CLK+	LVDS Channel 2, Clock +	
12	CH1_0-	LVDS Channel 1, Signal 0-		37	GND	Ground	
13	CH1_0+	LVDS Channel 1, Signal 0+		38	CH2_3-	LVDS Channel 2, Signal 3-	
14	CH1_1-	LVDS Channel 1, Signal 1-		39	CH2_3+	LVDS Channel 2, Signal 3+	
15	CH1_1+	LVDS Channel 1, Signal 1+		40	CH2_4-	LVDS Channel 2, Signal 4-	
16	CH1_2-	LVDS Channel 1, Signal 2-		41	CH2_4+	LVDS Channel 2, Signal 4+	
17	CH1_2+	LVDS Channel 1, Signal 2+		42	GND	Ground	
18	GND	Ground		43	N.C.	No connection	2
19	CH1_CLK-	LVDS Channel 1, Clock -		44	GND	Ground	
20	CH1_CLK+	LVDS Channel 1, Clock +		45	GND	Ground	
21	GND	Ground		46	GND	Ground	
22	CH1_3-	LVDS Channel 1, Signal 3-		47	N.C.	No connection	2
23	CH1_3+	LVDS Channel 1, Signal 3+		48	V _{DD}	Power Supply Input Voltage	
24	CH1_4-	LVDS Channel 1, Signal 4-		49	V _{DD}	Power Supply Input Voltage	
25	CH1_4+	LVDS Channel 1, Signal 4+		50	V _{DD}	Power Supply Input Voltage	
				51	V _{DD}	Power Supply Input Voltage	

Note1. Pin number start from the left side as the following figure.



Note2. Please leave this pin unoccupied. It can not be connected by any signal (Low/GND/High).

Note3. Input control signal threshold voltage definition

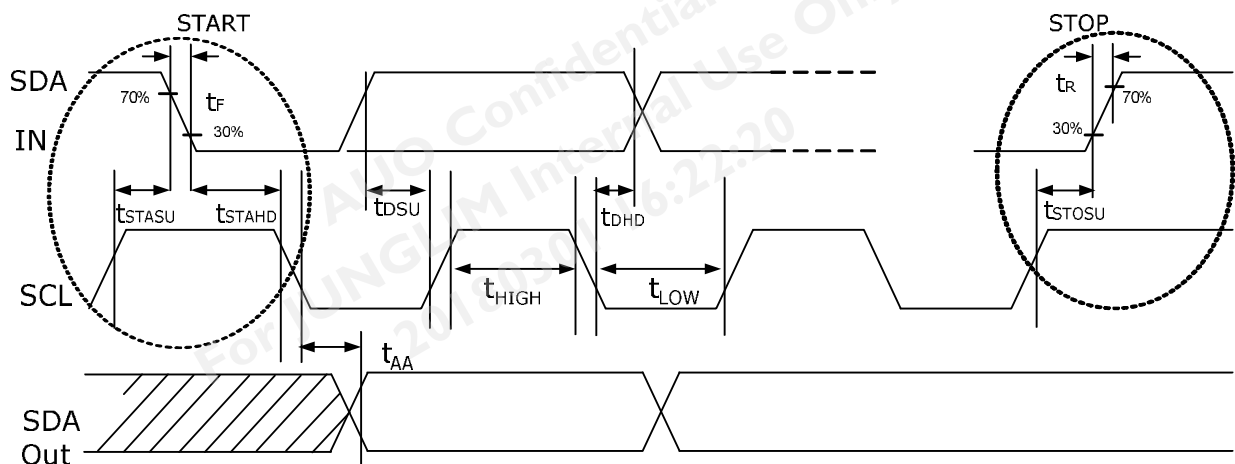
Item	Symbol	Min.	Typ.	Max.	Unit
Input High Threshold Voltage	VIH	2.7	-	3.6	V
Input Low Threshold Voltage	VIL	0	-	0.6	V

Note4. I2C Data and Clock

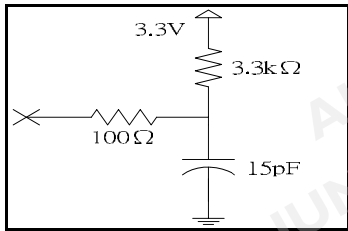
I2C Data and Clock timing

Parameter		Symbol	Min.	Typ.	Max	Unit
I2C	SCL clock frequency	fSCL	-	-	350	kHz
	Clock Pulse Width Low	tLOW	1.85	-	-	us
	Clock Pulse Width High	tHIGH	0.4	-	-	us
	Clock Low to Data Output Valid	tAA	1.76	-	-	us
	Start Setup Time	tSTASU	0.6	-	-	us
	Start Hold Time	tSTAHD	0.6	-	-	us
	Stop Setup Time	tSTOSU	0.6	-	-	us
	Data In Setup Time	tDSU	0.1	-	-	us
	Data In Hold Time	tDHD	0	-	-	us
	SCL/SDA Rise Time	tR	-	-	0.3	us
	SCL/SDA Fall Time	tF	-	-	0.3	us

I2C Read/Write Timing



Input equivalent impedance of SDA/SCL pin

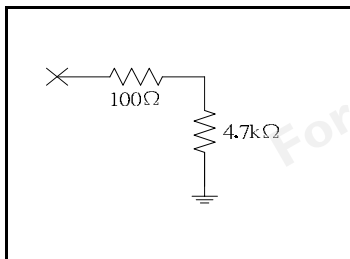


Note5. Write Protection

Mode selection

WP	Note
L or OPEN	Protection
H	Writable

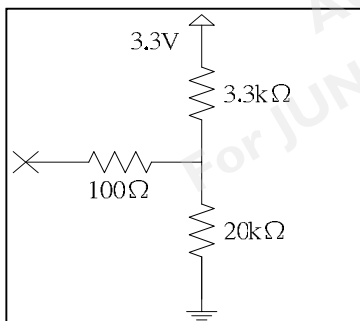
Input equivalent impedance of WP pin



Note6. LVDS data format selection

LVDS_SEL	Mode
H or OPEN	NS
L	Jeida

Input equivalent impedance of LVDS_SEL pin

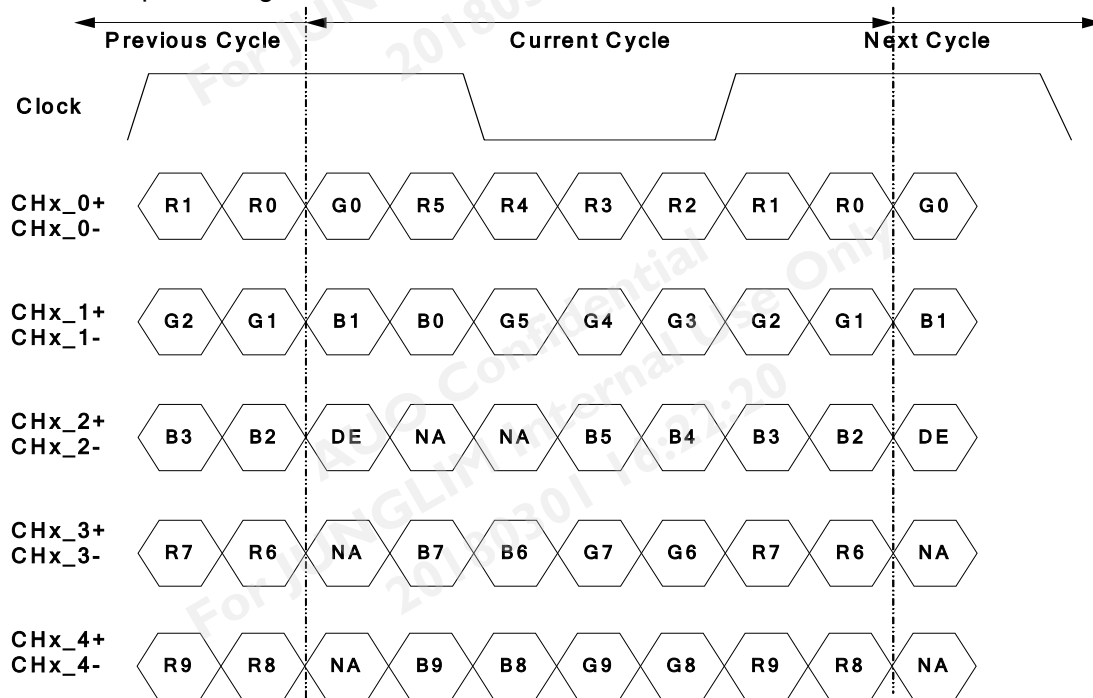


3.3. Input Data Format

3.3.1. Data mapping

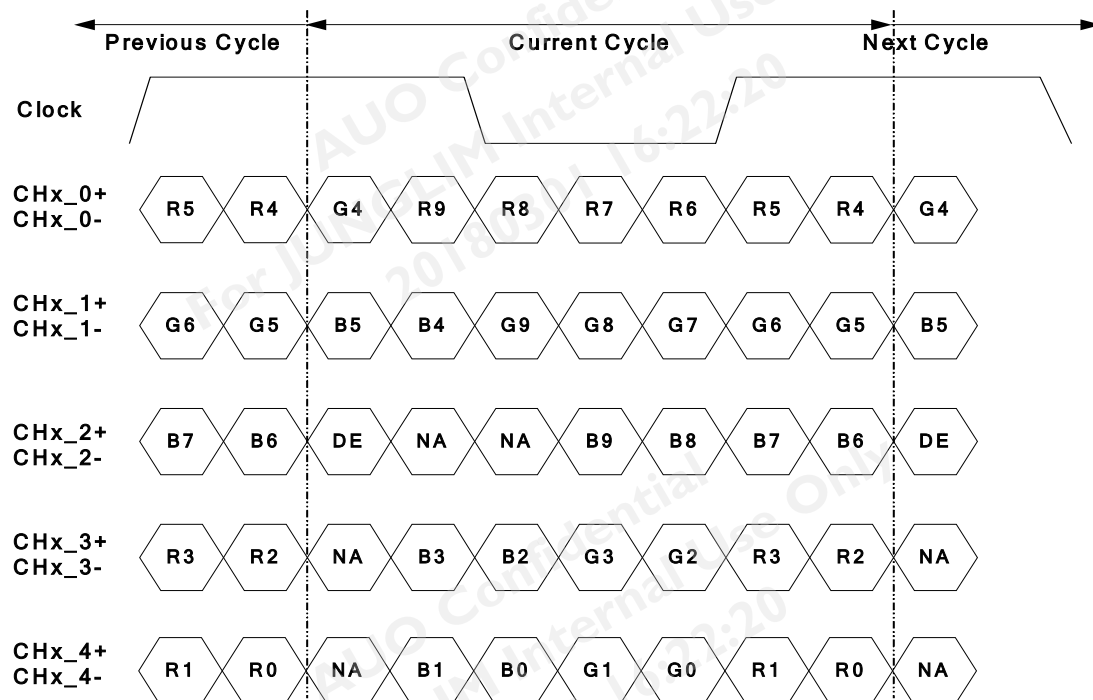
LVDS Option for 10bit

- LVDS Option = High/OPEN → NS



Note: x = 1, 2, 3, 4...

- LVDS Option = Low → JEIDA



Note: x = 1, 2, 3, 4...

The brightness of each primary color (red, green and blue) is based on the 10 bit gray scale data input for the color; the higher the binary input, the brighter the color. The table below provides a reference for color versus data input.

[illegible]

3.4. Signal Timing Specification

This is the signal timing required at the input of the user connector. All of the interface signal timing should be satisfied with the following specifications for its proper operation.

Timing Table (DE only Mode)

Signal	Item	Symbol	Min.	Typ.	Max	Unit
Vertical Section	Period	Tv	1100	1125	1480	Th
	Active	Tdisp (v)	1080			
	Blanking	Tblk (v)	20	45	400	Th
Horizontal Section	Period	Th	1030	1100	1325	Tclk
	Active	Tdisp (h)	960			
	Blanking	Tblk (h)	70	140	365	Tclk
Clock	Frequency	Fclk=1/Tclk	53	74.25	82	MHz
Vertical Frequency	Frequency	Fv	47	60	63	Hz
Horizontal Frequency	Frequency	Fh	60	67.5	73	KHz

Notes:

(1) Display position is specific by the rise of DE signal only.

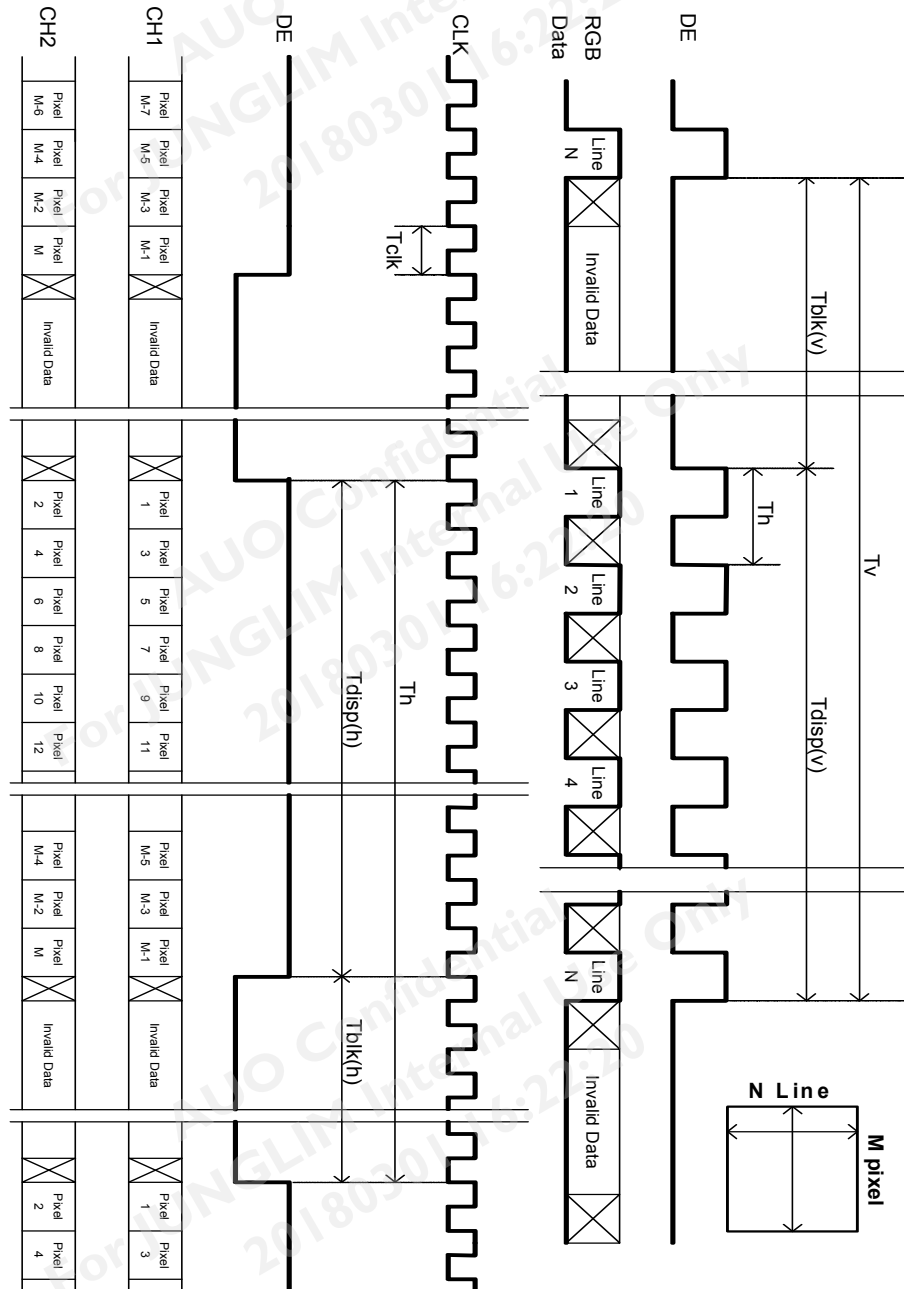
Horizontal display position is specified by the rising edge of 1st DCLK after the rise of 1st DE, is displayed on the left edge of the screen.

(2) Vertical display position is specified by the rise of DE after a “Low” level period equivalent to eight times of horizontal period. The 1st data corresponding to one horizontal line after the rise of 1st DE is displayed at the top line of screen.

(3) If a period of DE “High” is less than 1920 DCLK or less than 1080 lines, the rest of the screen displays black.

(4) The display position does not fit to the screen if a period of DE “High” and the effective data period do not synchronize with each other.

Signal Timing Waveforms



Note1. Display position is specific by the rise of DE signal only.

Horizontal display position is specified by the rising edge of 1st DCLK after the rise of 1st DE, is displayed on the left edge of the screen.

Note2. Vertical display position is specified by the rise of DE after a "Low" level period equivalent to eight times of horizontal period. The 1st data corresponding to one horizontal line after the rise of 1st DE is displayed at the top line of screen

Note3. If a period of DE "High" is less than 3840 DCLK or less than 2160 lines, the rest of the screen displays black.

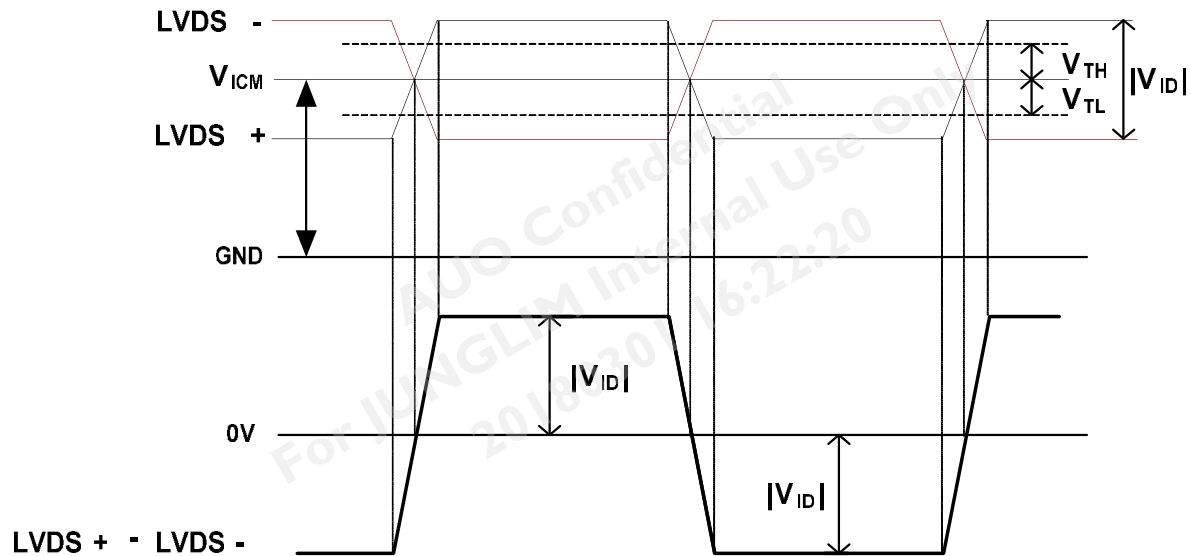
Note4. The display position does not fit to the screen if a period of DE "High" and the effective data period do not synchronize with each other.

3.5. Input interface characteristics

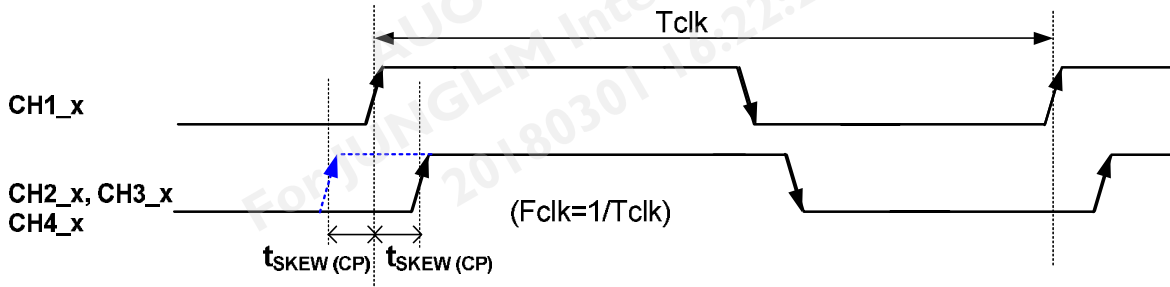
3.5.1. LVDS

Parameter		Symbol	Value			Unit	Note
			Min.	Typ.	Max		
LVDS Interface	Input Differential Voltage	$ V_{ID} $	200	400	600	mV _{DC}	1
	Differential Input High Threshold Voltage	V_{TH}	+100	--	+300	mV _{DC}	1
	Differential Input Low Threshold Voltage	V_{TL}	-300	--	-100	mV _{DC}	1
	Input Common Mode Voltage	V_{ICM}	1.1	1.25	1.4	V _{DC}	1
	Input Channel Pair Skew Margin	$t_{SKEW (CP)}$	-500	--	+500	ps	2
	Input Channel Pair Skew Margin (only for M'Star MST7428BB)	$t_{SKEW (CP)}$	-400	--	+400	ps	2
	Receiver Clock : Spread Spectrum Modulation range	Fclk_ss	Fclk -3%	--	Fclk +3%	MHz	3
	Receiver Clock : Spread Spectrum Modulation frequency	Fss	30	--	200	KHz	3
	Receiver Data Input Margin Fclk = 85 MHz Fclk = 65 MHz	tRMG	-0.4 -0.5	-- --	0.4 0.5	ns	4

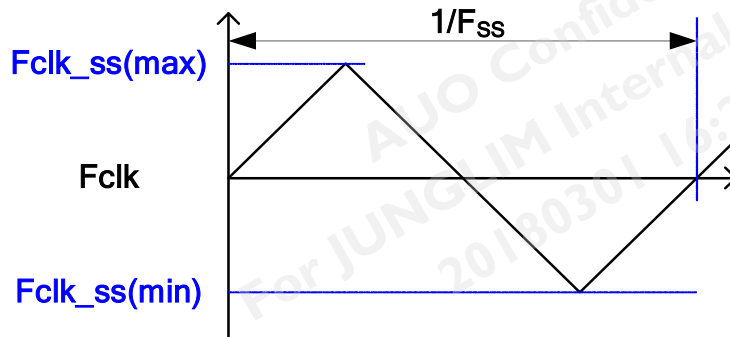
Note1. $V_{ICM} = 1.25V$



Note2. Input Channel Pair Skew Margin

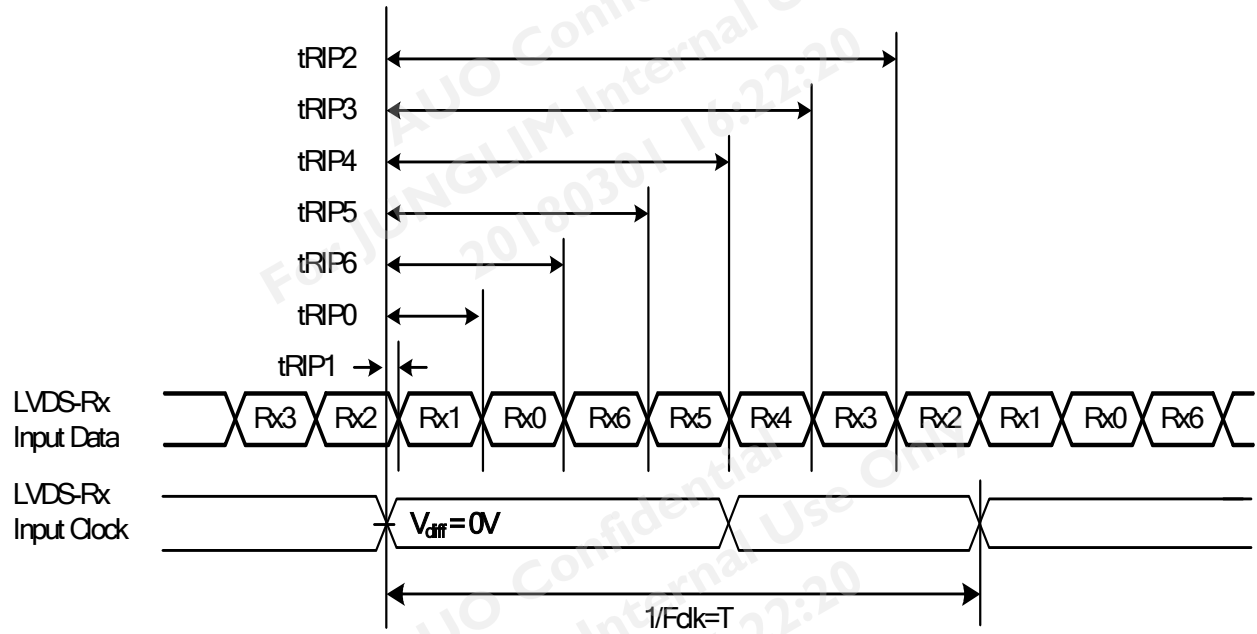


Note3. LVDS Receiver Clock SSCG (Spread spectrum clock generator) is defined as below figures.

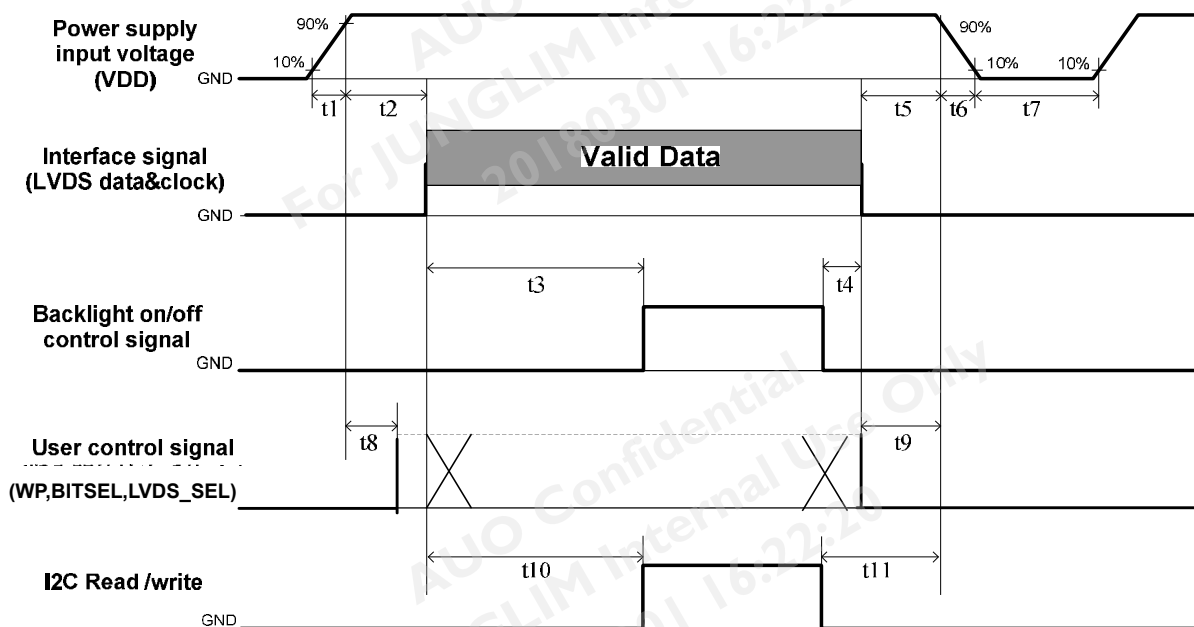


Note4. Receiver Data Input Margin

Parameter	Symbol	Rating			Unit	Note
		Min	Type	Max		
Input Clock Frequency	Fclk	Fclk (min)	--	Fclk (max)	MHz	$T=1/F_{clk}$
Input Data Position0	tRIP1	- tRMG	0	tRMG	ns	
Input Data Position1	tRIP0	$T/7 - tRMG $	$T/7$	$T/7 + tRMG $	ns	
Input Data Position2	tRIP6	$2T/7 - tRMG $	$2T/7$	$2T/7 + tRMG $	ns	
Input Data Position3	tRIP5	$3T/7 - tRMG $	$3T/7$	$3T/7 + tRMG $	ns	
Input Data Position4	tRIP4	$4T/7 - tRMG $	$4T/7$	$4T/7 + tRMG $	ns	
Input Data Position5	tRIP3	$5T/7 - tRMG $	$5T/7$	$5T/7 + tRMG $	ns	
Input Data Position6	tRIP2	$6T/7 - tRMG $	$6T/7$	$6T/7 + tRMG $	ns	



3.6. Power Sequence for LCD



Parameter	Values			Unit
	Min.	Type.	Max.	
t1	0.4	---	30	ms
t2	0.1	---	50	ms
t3	400	---	---	ms
t4	0 ^{*1}	---	---	ms
t5	0	---	---	ms
t6	---	---	--- ^{*2}	ms
t7	1000 ^{*3}	---	---	ms
t8	20 ^{*4}	---	50	ms
t9	0	---	---	ms
t10	400	---	---	ms
t11	150	---	---	ms

Note:

- (1) t4=0 : concern for residual pattern before BLU turn off.
- (2) t6 : voltage of VDD must decay smoothly after power-off. (customer system decide this value)
- (3) t7 : When the power supply input voltage(VDD) is off, be sure to pull down the valid and invalid data to 0V.
- (4) When user control signal is N.C. (no connection), opened in Transmitted end, t8 timing spec can be negligible.

3.7. Backlight Specification

3.7.1. Electrical specification ($T_a = 25 \pm 2 \text{ }^{\circ}\text{C}$)

	Item	Symbol	Condition	Spec			Unit	Note
				Min	Typ	Max		
1	Input Voltage	V _{DDB}	-	22.8	24	25.2	VDC	-
2	Input Current	I _{DDB}	V _{DDB} =24V		2.54	2.76	ADC	1
3	Input Power	P _{DDB}	V _{DDB} =24V		61	66.3	W	1
4	Inrush Current	I _{RUSH}	V _{DDB} =24V			4	ADC	2
5	On/Off control voltage	V _{B_{LON}}	ON	V _{DDB} =24V	2	-	VDC	-
			OFF		0	-		-
6	On/Off control current	I _{B_{LON}}	V _{DDB} =24V	-	-	1.5	mA	-
7	External PWM Control Voltage	V _{EPWM}	MAX	V _{DDB} =24V	2	-	VDC	-
			MIN	V _{DDB} =24V	0	-		-
8	External PWM Control Current	I _{EPWM}	V _{DDB} =24V	-	-	2	mADC	-
9	External PWM Duty ratio	D _{EPWM}	V _{DDB} =24V	10	-	100	%	3
10	External PWM Frequency	F _{EPWM}	V _{DDB} =24V	90	20K	50K	Hz	-
11	DET status signal	DET	HI	V _{DDB} =24V	Open Collector		VDC	4
			Lo		0	-	VDC	4
12	Input Impedance	R _{in}	V _{DDB} =24V	300			Kohm	-

Note 1 : Dimming ratio= 100% (MAX) ($T_a=25\pm5^{\circ}\text{C}$, Turn on for 45minutes)

Note 2: Measurement condition Rising time = 20ms (V_{DDB} : 10%~90%);

Note 3: Dimming linearity only guarantee between 20 ~ 80% duty. Less than 10% dimming control is functional well and no backlight shutdown happened

Note 4: Normal : 0~0.8V ; Abnormal : Open collector

3.7.2. Input Pin Assignment

LED driver board connector:

CN1: S14B-PHA-SM3-TB (HF) (JST)

Pin No	Symbol	Description
1	VDDDB	Operatng Voltage Supply, +24V DC regulated
2	VDDDB	Operatng Voltage Supply, +24V DC regulated
3	VDDDB	Operatng Voltage Supply, +24V DC regulated
4	VDDDB	Operatng Voltage Supply, +24V DC regulated
5	VDDDB	Operatng Voltage Supply, +24V DC regulated
6	BLGND	Ground and Current Return
7	BLGND	Ground and Current Return
8	BLGND	Ground and Current Return
9	BLGND	Ground and Current Return
10	BLGND	Ground and Current Return
11	DET	BLU status detection: Normal: 0~0.8V; Abnormal: Open collector (Recommend Pull high R > 10K, VDD=3.3V)
12	VBLOn	BL On-Off: High/Open (2.0V~5.5V) for BL On, Low (GND) for off
13	NC	NA
14	External PWM(PDIM)	External PWM (0%~100%)

CN2: CN2: CI0112M1HRD-NH (Cvilux)

Pin No	Symbol	Description
1	VLED1 +	LB1+ High Voltage
2	VLED1 +	LB1+ High Voltage
3	EB1-1 -	LB1-string1 feedback
4	EB1-2 -	LB1-string2 feedback
5	EB1-3 -	LB1-string3 feedback
6	EB1-4 -	LB1-string4 feedback
7	FB2-4 -	LB2-string4 feedback
8	FB2-3 -	LB2-string3 feedback

9	FB2-2 -	LB2-string2 feedback
10	FB2-1 -	LB2-string1 feedback
11	VLED2 +	LB2+ High Voltage
12	VLED2 +	LB2+ High Voltage

Note1. DET status

DET	BLU status
0 ~ 0.8V	Normal
Open collector	Abnormal

Recommend pull high R > 10K ohm, pull high voltage VDD = 3.3V

Note2. Input control signal threshold voltage definition

Item	Symbol	Min.	Typ.	Max.	Unit
Input High Threshold Voltage	VIH	2	-	5.5	V
Input Low Threshold Voltage	VIL	0	-	0.8	V

Note3. VBLON

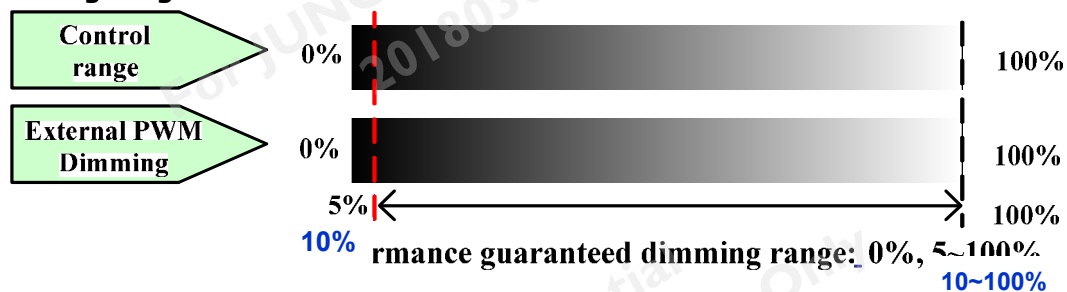
Mode selection

WP	Note
H or OPEN	BL On
L	BL Off

Note4. Please leave this pin unoccupied. It can not be connected by any signal (Low/GND/High).

Note5. PDIM

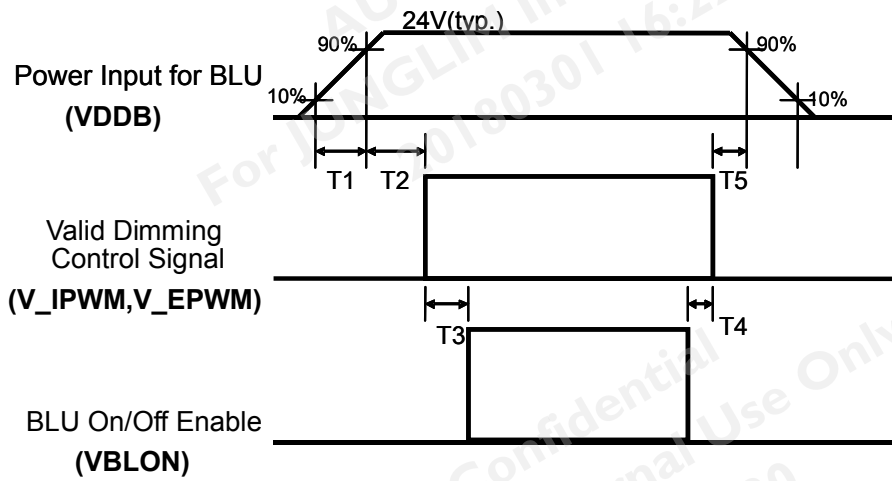
PWM Dimming range:



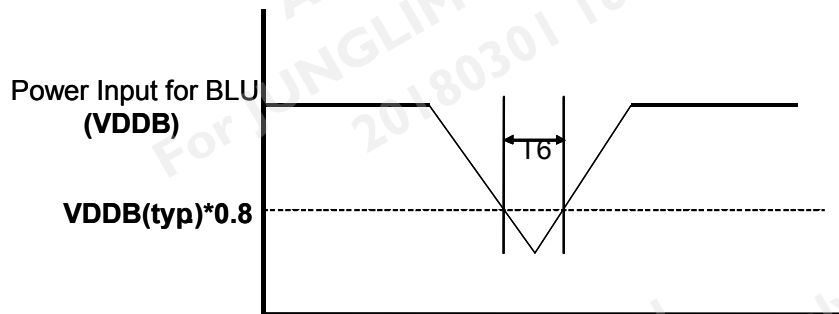
IF External PWM function less than 10% dimming ratio, Judge condition as below:

- (1) Backlight module must be lighted ON normally.
- (2) All protection function must work normally.
- (3) Uniformity and flicker could not be guaranteed

3.7.3. Power Sequence for Backlight



Dip condition



Parameter	Value			Units
	Min	Typ	Max	
T1	20	-	-	ms
T2	250	-	-	ms
T3	200	-	-	ms
T4	0	-	-	ms
T5	0	-	-	ms
T6	-	-	1000	ms ^{*1}

Note: 1. T1 describes rising time of 0V to 24V and this parameter does not applied at restarting time. Even though T1 is over the specified value, there is no problem if I2t spec of fuse is satisfied.

2. T8 describes VDDDB dip condition and VDDDB couldn't lower than 10% VDDDB.

3.7.4. LED Operating Life Time

Parameter	Symbol	Value			Unit	Note
		Min.	Typ.	Max		
Backlight Operating Life Time(MTTF)		30000			Hour	1

Note:

- The lifetime (MTTF) is defined as the time which luminance of LED is 50% compared to its original value.
 [Operating condition: Continuous operating at $T_a = 25 \pm 2^\circ\text{C}$, for single lamp/LED only]

4. Reliability Test Items

	Test Item	Q'ty	Condition
1	High temperature storage test	3	60°C, 500hrs
2	Low temperature storage test	3	-20°C, 500hrs
3	High temperature operation test	3	50°C, 500hrs
4	Low temperature operation test	3	-5°C, 500hrs
5	Vibration test (With carton)	1(PKG)	Random wave (1.04Grms 2~200Hz) Duration : X,Y,Z 20min per axes
6	Drop test (With carton)	1(PKG)	Surround four flats drop height:15 cm Bottom flat drop height:25.4 cm twice (ASTMD4169)

5. International Standard

5.1. Safety

- (1) UL 60950-1; Standard for Safety of Information Technology Equipment Including electrical Business Equipment.
- (2) IEC 60950-1; Standard for Safety of International Electrotechnical Commission
- (3) EN 60950-1; European Committee for Electrotechnical Standardization (CENELEC), EUROPEAN STANDARD for Safety of Information Technology Equipment Including Electrical Business Equipment.

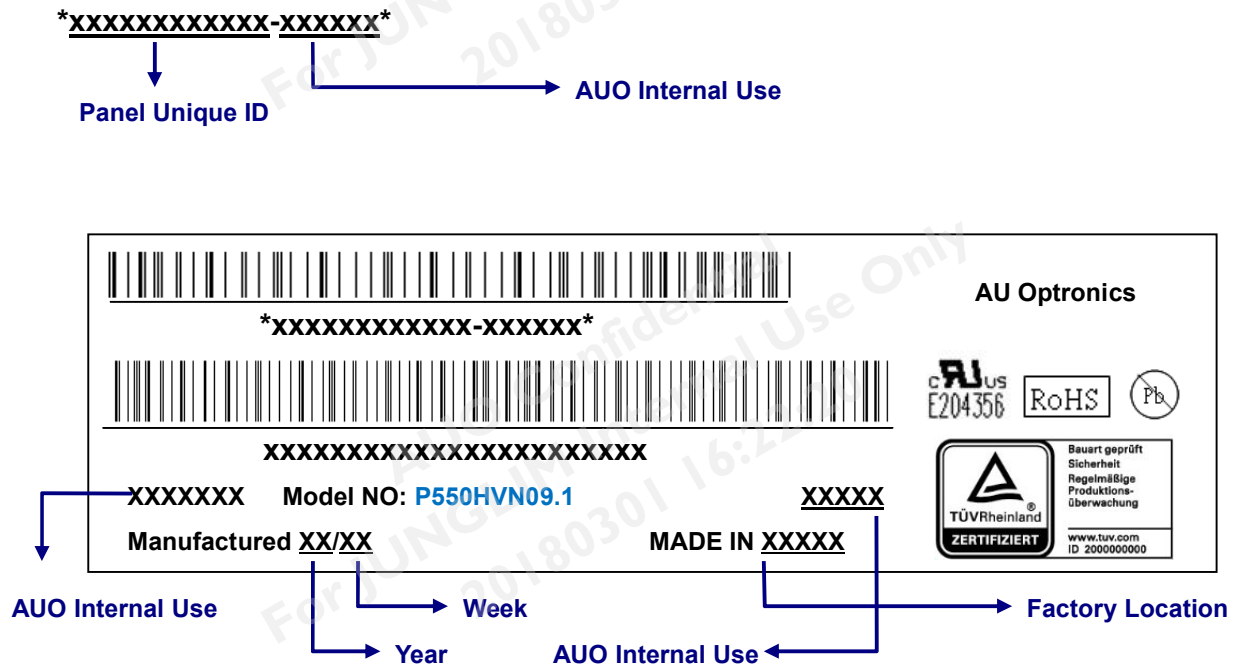
5.2. EMC

- (1) ANSI C63.4 "Methods of Measurement of Radio-Noise Emissions from Low-Voltage Electrical and Electrical Equipment in the Range of 9kHz to 40GHz. "American National standards Institute(ANSI), 1992
- (2) C.I.S.P.R "Limits and Methods of Measurement of Radio Interface Characteristics of Information Technology Equipment." International Special committee on Radio Interference.
- (3) EN 55022 "Limits and Methods of Measurement of Radio Interface Characteristics of Information Technology Equipment." European Committee for Electrotechnical Standardization. (CENELEC), 1998

6. Packing

6.1. Definition of Label

A. Panel Label:



Green mark description

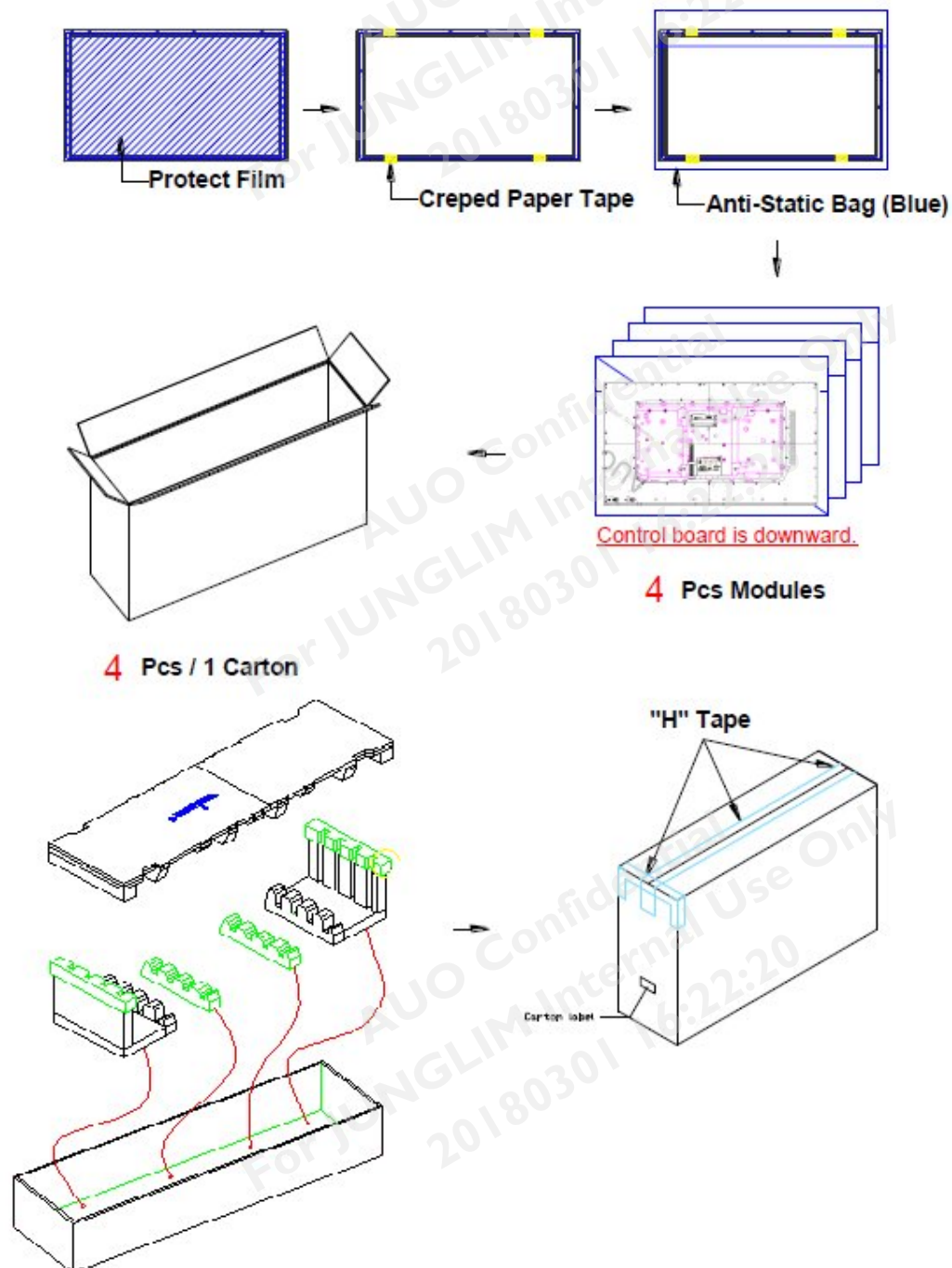
- (1) For Pb Free Product, AUO will add  for identification.
- (2) For RoHS compatible products, AUO will add **RoHS** for identification.

Note: The green Mark will be present only when the green documents have been ready by AUO internal green team. (definition of green design follows the AUO green design checklist.)

B. Carton Label:

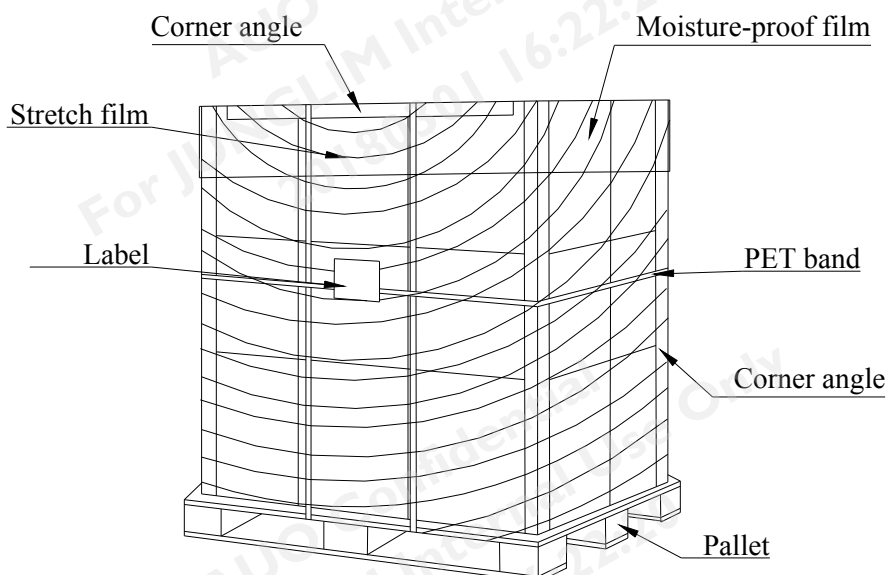
AU Optronics	QTY: 4	RoHS	
MODEL NO: P550HVN09.1			
PART NO: 97.55P16.101			
CUSTOMER NO:			
CARTON NO:			
Made in XXXXXX	*XXXXXX-XXXXXXXXXX*		

6.2. Packing Methods



6.3. Pallet and Shipment Information

	Item	Specification			Packing Remark
		Qty.	Dimension	Weight (kg)	
1	Packing Box	4 pcs/box	1305(L)*383(W)*800(H)	68.1	
2	Pallet	1	1315(L)*1150(W)*138(H)	17	
3	Boxes per Pallet	3 boxes/pallet			
4	Panels per Pallet	12pcs/pallet			
5	Pallet after packing	12	1315(L)*1150(W)*938(H)	221.3	



7. Precautions

Please pay attention to the followings when you use this TFT LCD module.

7.1. Mounting Precautions

- (1) You must mount a module using holes arranged in four corners or four sides.
- (2) You should consider the mounting structure so that uneven force (ex. twisted stress) is not applied to module. And the case on which a module is mounted should have sufficient strength so that external force is not transmitted directly to the module.
- (3) Please attach the surface transparent protective plate to the surface in order to protect the polarizer. Transparent protective plate should have sufficient strength in order to resist external force.
- (4) You should adopt radiation structure to satisfy the temperature specification.
- (5) Acetic acid type and chlorine type materials for the cover case are not desirable because the former generates corrosive gas of attacking the polarizer at high temperature and the latter cause circuit broken by electro-chemical reaction.
- (6) Do not touch, push or rub the exposed polarizer with glass, tweezers or anything harder than HB pencil lead. And please do not rub with dust clothes with chemical treatment. Do not touch the surface of polarizer for bare hand or greasy cloth. (Some cosmetics are detrimental to the polarizer.)
- (7) When the surface becomes dusty, please wipe gently with absorbent cotton or other soft materials like chamois soaks with petroleum benzene. Normal-hexane is recommended for cleaning the adhesives used to attach front/ rear polarizer. Do not use acetone, toluene and alcohol because they cause chemical damage to the polarizer.
- (8) Wipe off saliva or water drops as soon as possible. Their long time contact with polarizer causes deformations and color fading.
- (9) Do not open the case because inside circuits do not have sufficient strength.

7.2. Operating Precautions

- (1) The spike noise causes the mis-operation of circuits. It should be lower than following voltage:
 $V=\pm 200\text{mV}$ (Over and under shoot voltage)
- (2) Response time depends on the temperature. (In lower temperature, it becomes longer..)
- (3) Brightness depends on the temperature. (In lower temperature, it may become lower.) And in lower temperature, response time (required time that brightness is stable after turned on) becomes longer.
- (4) Be careful for condensation at sudden temperature change. Condensation makes damage to polarizer or electrical contacted parts. And after fading condensation, smear or spot will occur.
- (5) When fixed patterns are displayed for a long time, remnant image is likely to occur.
- (6) Module has high frequency circuits. Sufficient suppression to the electromagnetic interference shall be done by system manufacturers. Grounding and shielding methods may

be important to minimize the interface.

7.3. Operating Condition for Public Information Display

The device listed in the product specification is designed and manufactured for PID (Public Information Display) application. To optimize module's lifetime and function, below operating usages are required.

(1) Normal operating condition

- A. Operating temperature: 0~40℃
- B. Operating humidity: 10~90%
- C. Display pattern: dynamic pattern (Real display).

Note) Long-term static display would cause image sticking.

(1) Operation usage to protect against image sticking due to long-term static display.

- A. Suitable operating time: 24 hours a day or less.
(* The moving picture can be allowed for 24 hours a day)
- B. Liquid Crystal refresh time is required. Cycling display between 5 minutes' information (static) display and 10 seconds' moving image.
- C. Periodically change background and character (image) color.
- D. Avoid combination of background and character with large different luminance.

(2) Periodically adopt one of the following actions after long time display.

- A. Running the screen saver (motion picture or black pattern)
- B. Power off the system for a while

(3) LCD system is required to place in well-ventilated environment. Adapting active cooling system is highly recommended.

(4) Product reliability and functions are only guaranteed when the product is used under right operation usages. If product will be used in extreme conditions, such as high temperature/humidity, display stationary patterns, or long operation time etc..., it is strongly recommended to contact AUO for filed application engineering advice. Otherwise, its reliability and function may not be guaranteed. Extreme conditions are commonly found at airports, transit stations, banks, stock market and controlling systems.

7.4. Electrostatic Discharge Control

Since a module is composed of electronic circuits, it is not strong to electrostatic discharge. Make certain that treatment persons are connected to ground through wristband etc. And don't touch interface pin directly.

7.5. Precautions for Strong Light Exposure

Strong light exposure causes degradation of polarizer and color filter.

7.6. Storage

When storing modules as spares for a long time, the following precautions are necessary.

- (1) Store them in a dark place. Do not expose the module to sunlight or fluorescent light. Keep the temperature between 5°C and 35°C at normal humidity.
- (2) The polarizer surface should not come in contact with any other object. It is recommended that they be stored in the container in which they were shipped.
- (3) Storage condition is guaranteed under packing conditions.
- (4) The phase transition of Liquid Crystal in the condition of the low or high storage temperature will be recovered when the LCD module returns to the normal condition.

7.7. Handling Precautions for Protection Film

- (1) The protection film is attached to the bezel with a small masking tape. When the protection film is peeled off, static electricity is generated between the film and polarizer. This should be peeled off slowly and carefully by people who are electrically grounded and with well ion-blown equipment or in such a condition, etc.
- (2) When the module with protection film attached is stored for a long time, sometimes there remains a very small amount of glue still on the bezel after the protection film is peeled off.
- (3) You can remove the glue easily. When the glue remains on the bezel or its vestige is recognized, please wipe them off with absorbent cotton waste or other soft material like chamois soaked with normal-hexane.